

# **SIMATS ENGINEERING**

## **ASSESSMENT TOOL 1**

**COURSE NAME:** Artificial Intelligence      **COURSE CODE:** CSA17

### **COURSE OUTCOME COVERED**

**CO3:** *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

*Assessment Tool Weightage: 50%*

**Time Duration: 1 Hour**

**QUESTIONS: 4**

**Total Marks:40**

Annexure A

### **CASE STUDY**

#### **Code of Conduct:**

*I Karthikeyan s (192419048) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

*I understand that any violation of academic integrity rules will result in disciplinary action.*

**Signature of the student: S.Karthikeyan**

## CASE STUDY 1 – SMART MEDICAL DIAGNOSIS SYSTEM

### (a) Representation using Propositional Logic

The knowledge base contains three rules and four facts. Let the following propositional symbols be used:

| Symbol | Meaning                    |
|--------|----------------------------|
| F      | Patient has Fever          |
| C      | Patient has Cough          |
| R      | Patient has Rash           |
| B      | Patient has Breathlessness |
| Flu    | Patient may have Flu       |
| M      | Patient may have Measles   |
| P      | Patient may have Pneumonia |

The rules can be represented precisely as follows:

**Rule 1:**  $(F \wedge C) \rightarrow \text{Flu}$

**Rule 2:**  $(F \wedge R) \rightarrow M$

**Rule 3:**  $(C \wedge B) \rightarrow P$

For Patient A, the given facts are:

**F(A) = True, C(A) = True, B(A) = True**

There is no fact stating R(A). Therefore Rash(A) is not available in the initial knowledge base.

### (b) Forward Chaining – Derivation of All Possible Diagnoses

Forward chaining starts with known facts and repeatedly applies every rule whose antecedent is satisfied. The process continues until no new fact can be derived.

1. Initial knowledge: Fever(A), Cough(A), and Breathlessness(A).

2. Apply Rule 1. Its antecedent is Fever(A)  $\wedge$  Cough(A). Both facts are true, so Flu(A) is derived.

**Fever(A)  $\wedge$  Cough(A)  $\Rightarrow$  Flu(A)**

3. Apply Rule 2. It requires Fever(A)  $\wedge$  Rash(A). Although Fever(A) is true, Rash(A) is not given. Therefore Rule 2 cannot fire.

4. Apply Rule 3. Its antecedent is Cough(A)  $\wedge$  Breathlessness(A). Both are true, so Pneumonia(A) is derived.

**Cough(A)  $\wedge$  Breathlessness(A)  $\Rightarrow$  Pneumonia(A)**

5. Check the knowledge base again. No additional rule can be fired because Rash(A) is still unavailable.

Final forward-chaining closure:

**{Fever(A), Cough(A), Breathlessness(A), Flu(A), Pneumonia(A)}**

Therefore, the possible diagnoses supported by the knowledge base are Flu and Pneumonia. Measles is not concluded.

### (c) Backward Chaining – Verify Pneumonia(A)

Backward chaining starts with the goal and works backwards to find rules that could establish that goal.

**Goal: Pneumonia(A)**

Only Rule 3 concludes Pneumonia:

**(Cough  $\wedge$  Breathlessness)  $\rightarrow$  Pneumonia**

Therefore the goal is reduced to two sub-goals:

**Pneumonia(A)  $\leftarrow$  Cough(A)  $\wedge$  Breathlessness(A)**

1. Check Cough(A): it is a given fact, so the first sub-goal succeeds.

2. Check Breathlessness(A): it is also a given fact, so the second sub-goal succeeds.

3. Since both sub-goals are satisfied, the original goal Pneumonia(A) is proved.

**$\therefore$  Pneumonia(A) is TRUE**

Hence, forward chaining and backward chaining both confirm that Patient A may have Pneumonia.

## CASE STUDY 2 – AUTONOMOUS TRAFFIC MANAGEMENT AGENT

### (a) Unification of Rule 1 with the Given Facts

Rule 1 is:

$\forall x [\text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)]$

The knowledge base contains:

**Vehicle(Ambulance)**

**EmergencyType(Ambulance)**

The variable x in Rule 1 must be replaced by Ambulance so that the predicates match the facts.

$\theta = \{x / \text{Ambulance}\}$

Applying the substitution  $\theta$  gives:

**Vehicle(Ambulance)  $\wedge$  EmergencyType(Ambulance)  $\rightarrow$  ClearPath(Ambulance)**

1. Match Vehicle(x) with Vehicle(Ambulance):  $x = \text{Ambulance}$ .
2. Match EmergencyType(x) with EmergencyType(Ambulance):  $x = \text{Ambulance}$ .
3. Both substitutions agree, so the substitution set is consistent and is the MGU for this matching.
4. Since both antecedents are facts, ClearPath(Ambulance) is derived.  
 $\therefore$  **ClearPath(Ambulance)**

### (b) Resolution Proof for Proceed(CarA)

The relevant rules are converted into clauses by eliminating implications.

**Rule 2:  $\neg \text{ClearPath}(x) \vee \text{GreenSignal}(x)$**

**Rule 3:  $\neg \text{Vehicle}(x) \vee \neg \text{Behind}(x,y) \vee \neg \text{GreenSignal}(y) \vee \text{Proceed}(x)$**

Given facts:

**Vehicle(CarA), Behind(CarA,Ambulance)**

**Vehicle(Ambulance), EmergencyType(Ambulance)**

Resolution derivation:

1. From Vehicle(Ambulance) and EmergencyType(Ambulance), Rule 1 gives ClearPath(Ambulance).

**Vehicle(Ambulance)  $\wedge$  EmergencyType(Ambulance)  $\rightarrow$  ClearPath(Ambulance)**

2. Instantiate Rule 2 with  $x = \text{Ambulance}$ :

**$\neg \text{ClearPath}(\text{Ambulance}) \vee \text{GreenSignal}(\text{Ambulance})$**

3. Resolve the clause above with ClearPath(Ambulance). The complementary literals cancel, giving:

**GreenSignal(Ambulance)**

4. Instantiate Rule 3 with  $x = \text{CarA}$  and  $y = \text{Ambulance}$ :

**$\neg \text{Vehicle}(\text{CarA}) \vee \neg \text{Behind}(\text{CarA}, \text{Ambulance}) \vee \neg \text{GreenSignal}(\text{Ambulance}) \vee \text{Proceed}(\text{CarA})$**

5. Resolve with the facts Vehicle(CarA), Behind(CarA,Ambulance), and GreenSignal(Ambulance). The remaining literal is:  
**Proceed(CarA)**

Therefore, by resolution:

$\therefore$  **Proceed(CarA) is TRUE**

### (c) Ability to Handle Two Simultaneous Emergency Vehicles

The current knowledge base can reason about an emergency vehicle individually. Rule 1 can produce ClearPath for each emergency vehicle, and Rule 2 can produce a green signal. However, the knowledge base does not contain explicit reasoning about conflicts between two emergency vehicles.

For example, if two ambulances require overlapping or conflicting paths, the present rules do not state which one has priority or how the controller should coordinate them.

Additional rules or facts should include:

- Priority(E1,E2): specifies that emergency vehicle E1 has higher priority than E2.
- Location and route facts for each emergency vehicle.
- A conflict rule identifying when two routes cannot be cleared at the same time.

## CASE STUDY 3 – AGRICULTURAL EXPERT REASONING SYSTEM

### (a) Conversion of P1, P2, P3 and Facts into CNF

P1:  $\text{SoilDry} \rightarrow \text{IrrigationNeeded}$

Eliminate the implication using  $A \rightarrow B \equiv \neg A \vee B$ :

$\neg \text{SoilDry} \vee \text{IrrigationNeeded}$

This is already a CNF clause.

P2:  $\text{IrrigationNeeded} \wedge \text{CropWheat} \rightarrow \text{ApplyDripMethod}$

$\neg(\text{IrrigationNeeded} \wedge \text{CropWheat}) \vee \text{ApplyDripMethod}$

Using De Morgan's law:

$\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}$

This is a single CNF clause.

P3:  $\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$

$\neg(\neg \text{ApplyDripMethod} \wedge \text{CropWheat}) \vee \text{CropAtRisk}$

Using De Morgan's law:

$\text{ApplyDripMethod} \vee \neg \text{CropWheat} \vee \text{CropAtRisk}$

The given facts are unit clauses:

**SoilDry**

**CropWheat**

### (b) Resolution Refutation – Prove ApplyDripMethod

To prove **ApplyDripMethod** by resolution refutation, first negate the goal and add the negated goal to the CNF knowledge base.

**Goal: ApplyDripMethod**

**Negated goal:  $\neg \text{ApplyDripMethod}$**

Relevant clauses:

**C1:  $\neg \text{SoilDry} \vee \text{IrrigationNeeded}$**

**C2:  $\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}$**

**C3: SoilDry**

**C4: CropWheat**

**C5:  $\neg \text{ApplyDripMethod}$**

Resolution steps:

1. Resolve C1 with C3. **SoilDry** and  $\neg \text{SoilDry}$  are complementary:

$\neg \text{SoilDry} \vee \text{IrrigationNeeded}, \text{SoilDry} \Rightarrow \text{IrrigationNeeded}$

2. Resolve the result with C2:

$\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}, \text{IrrigationNeeded}$

$\Rightarrow \neg \text{CropWheat} \vee \text{ApplyDripMethod}$

3. Resolve with C4, **CropWheat**:

$\neg \text{CropWheat} \vee \text{ApplyDripMethod}, \text{CropWheat} \Rightarrow \text{ApplyDripMethod}$

4. Resolve the derived **ApplyDripMethod** with the negated goal C5:

**ApplyDripMethod,  $\neg \text{ApplyDripMethod} \Rightarrow \square$**

The empty clause  $\square$  represents a contradiction. Therefore the negation of the goal is inconsistent with the knowledge base.

**$\therefore$  ApplyDripMethod is PROVED**

### (c) Effect of Removing SoilDry

1. If **SoilDry** is removed, Rule P1 cannot be applied because its antecedent is no longer known to be true.

2. Consequently, **IrrigationNeeded** cannot be derived from P1.

3. Without **IrrigationNeeded**, P2 cannot derive **ApplyDripMethod** from the remaining facts.

## CASE STUDY 4 – WUMPUS WORLD KNOWLEDGE AGENT

### (a) Belief State and Modus Ponens

The agent receives the following percepts:

**Stench[1,2], Breeze[1,1], Glitter[2,2]**

Using the rules supplied in the case study, the agent derives the following knowledge:

| Percept      | Rule   | Conclusion                   |
|--------------|--------|------------------------------|
| Stench[1,2]  | Rule 1 | Wumpus is adjacent to [1,2]. |
| Breeze[1,1]  | Rule 2 | Pit is adjacent to [1,1].    |
| Glitter[2,2] | Rule 3 | Gold is at [2,2].            |

1. Stench[1,2] is true. By Rule 1, the Wumpus must be in a cell adjacent to [1,2].
2. Breeze[1,1] is true. By Rule 2, a pit must be in a cell adjacent to [1,1].
3. Glitter[2,2] is true. By Rule 3, gold is located at [2,2].
4. Rule 4 can derive Safe[x,y] only when both  $\neg$ Wumpus[x,y] and  $\neg$ Pit[x,y] are known. Such negative facts are not supplied, so safety cannot be established for a particular cell using Rule 4 alone.

Thus, the belief state contains the three derived conclusions above, while exact safe-cell information remains undetermined.

### (b) Forward Chaining – Possible Wumpus and Pit Locations

For the Wumpus, Stench[1,2] means that the Wumpus is adjacent to [1,2]. Under the normal four-neighbour grid interpretation, the in-bound candidate cells are:

**[1,1], [1,3], [2,2]**

For the pit, Breeze[1,1] means that a pit is adjacent to [1,1]. The in-bound candidate cells are:

**[1,2], [2,1]**

Forward chaining therefore narrows the possibilities to these candidate locations, but the supplied rules do not contain enough negative information to select exactly one candidate.

The Glitter percept directly establishes that Gold[2,2] is true.

### (c) Safety of the Path [1,1] → [1,2] → [2,2]

The proposed path must be evaluated using only what can be logically established from the given knowledge base.

1. At [1,1], a Breeze is perceived. Therefore a pit is adjacent to [1,1]. One possible adjacent location is [1,2].
  2. At [1,2], a Stench is perceived. Therefore the Wumpus is adjacent to [1,2].
  3. The knowledge base does not establish  $\neg$ Pit[1,2] or  $\neg$ Wumpus[1,2]. Therefore [1,2] cannot be classified as safe using Rule 4.
  4. Similarly, the available information does not provide enough negative facts to prove that [2,2] is free from both hazards.
  5. Because the safety of the intermediate cell [1,2] is not logically guaranteed, the complete path cannot be declared safe.
- ∴ The path [1,1] → [1,2] → [2,2] is NOT PROVEN SAFE.**

The agent should require additional percepts or negative knowledge before treating the route as logically safe. The fact that gold is at [2,2] does not by itself prove that the route to the gold is safe.